

Recitation 3

ECON UN3412: Econometrics

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1 Sampling Distribution & Variance of the OLS Estimator

In Recitation 2, we showed that the Ordinary Least Squares (OLS) slope estimator $\hat{\beta}_1$ is **unbiased** ($\mathbb{E}[\hat{\beta}_1] = \beta_1$) and **consistent** ($\hat{\beta}_1 \xrightarrow{p} \beta_1$).

However, unbiasedness only tells us where the sampling distribution is centered across repeated hypothetical samples. To perform **statistical inference**—such as constructing confidence intervals and testing hypotheses like $H_0 : \beta_1 = 0$ —we must know the **spread** (variance) and the **asymptotic shape** of this sampling distribution.

1.1 The Regression Model & Least Squares Assumptions

Consider the bivariate linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n$$

where Y_i is the dependent variable, X_i is the regressor, and u_i is the error term collecting all other unobserved factors influencing Y_i .

Our analysis relies on the three core **Least Squares Assumptions (LSAs)**:

Least Squares Assumptions (LSA 1–3):

1. **LSA 1 (Conditional Mean Zero):**

$$\mathbb{E}[u_i | X_i] = 0 \implies \text{Corr}(X_i, u_i) = 0$$

The error term has a mean of zero conditional on X_i . This guarantees exogeneity (no omitted variable bias).

2. **LSA 2 (i.i.d. Sampling):**

$(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ are independently and identically distributed (i.i.d.)

This holds when data are collected via simple random sampling from the population of interest.

3. LSA 3 (Large Outliers are Rare):

$$\mathbb{E}[X_i^4] < \infty \quad \text{and} \quad \mathbb{E}[u_i^4] < \infty$$

Nonzero, finite fourth moments guarantee that extreme outliers do not dominate the sample moments, allowing us to invoke the Central Limit Theorem.

1.2 Recap: The OLS Estimation Error

Recall from Recitation 2 that the OLS formula for the slope is:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

Substituting $Y_i - \bar{Y} = \beta_1(X_i - \bar{X}) + (u_i - \bar{u})$ yields:

$$\hat{\beta}_1 = \beta_1 \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sum_{i=1}^n (X_i - \bar{X})^2} + \frac{\sum_{i=1}^n (X_i - \bar{X})(u_i - \bar{u})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

Because $\sum_{i=1}^n (X_i - \bar{X})\bar{u} = \bar{u} \sum_{i=1}^n (X_i - \bar{X}) = 0$, the estimation error simplifies to:

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})u_i}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

1.3 Step-by-Step Derivation of $\text{Var}(\hat{\beta}_1)$

To derive the variance of $\hat{\beta}_1$ in large samples, we follow the step-by-step method presented in class:

Step 1: Normalize Numerator and Denominator by Sample Size

Divide both the numerator and the denominator of the estimation error by the sample size n :

$$\hat{\beta}_1 - \beta_1 = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})u_i}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

Step 2: Define the Composite Error Term v_i

Define the random variable:

$$v_i \equiv (X_i - \bar{X})u_i$$

(In large samples, replacing sample averages with population means gives $v_i \approx (X_i - \mu_X)u_i$.)

The numerator is simply the sample average of these v_i terms, which we denote by $\bar{v}$:

$$\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i$$

Step 3: Large-Sample Approximation of the Denominator

Consider the denominator, which is the sample variance of X (denoted s_X^2):

$$s_X^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

By the **Law of Large Numbers (LLN)** and LSA 2 & 3, as n grows large:

$$s_X^2 \xrightarrow{p} \text{Var}(X_i) = \sigma_X^2$$

In large samples, s_X^2 has negligible sampling variability and converges to the **constant** population variance σ_X^2 .

Replacing the random denominator with this constant gives the large-sample approximation:

$$\hat{\beta}_1 - \beta_1 \approx \frac{\frac{1}{n} \sum_{i=1}^n v_i}{\sigma_X^2} = \frac{\bar{v}}{\sigma_X^2}$$

Step 4: Distribution and Variance of the Numerator $\bar{v}$

Now examine the properties of $\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i$:

1. **Mean of v_i :** By LSA 1 ($\mathbb{E}[u_i | X_i] = 0$) and the Law of Iterated Expectations:

$$\mathbb{E}[v_i] = \mathbb{E}[(X_i - \mu_X)u_i] = \mathbb{E}\left[(X_i - \mu_X) \underbrace{\mathbb{E}[u_i | X_i]}_{=0}\right] = 0$$

Therefore, $\mathbb{E}[\bar{v}] = 0$.

2. **Variance of v_i :** Denote the population variance of v_i by σ_v^2 :

$$\sigma_v^2 \equiv \text{Var}(v_i) = \text{Var}((X_i - \mu_X)u_i)$$

3. **Variance of the Sample Average $\bar{v}$:** Because (X_i, Y_i) are i.i.d. (LSA 2), the terms v_i are also i.i.d. The variance of a sample mean of i.i.d. random variables is the individual variance divided by n :

$$\sigma_{\bar{v}}^2 \equiv \text{Var}(\bar{v}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n v_i\right) = \frac{\sigma_v^2}{n}$$

4. **Central Limit Theorem (CLT):** By the Central Limit Theorem, the normalized average

$\bar{v}$ is asymptotically normally distributed:

$$\bar{v} \stackrel{a}{\sim} \mathcal{N}(0, \sigma_v^2) = \mathcal{N}\left(0, \frac{\sigma_v^2}{n}\right)$$

Step 5: Applying the Variance Scaling Property

We now apply the scaling property of variances ($\text{Var}(cZ) = c^2 \text{Var}(Z)$ for any constant c).

Since σ_X^2 in the denominator acts as a fixed constant in large samples:

$$\begin{aligned} \text{Var}(\hat{\beta}_1) &= \text{Var}(\hat{\beta}_1 - \beta_1) \\ &\approx \text{Var}\left(\frac{\bar{v}}{\sigma_X^2}\right) \\ &= \frac{1}{(\sigma_X^2)^2} \text{Var}(\bar{v}) \\ &= \frac{1}{(\sigma_X^2)^2} \sigma_v^2 \\ &= \frac{1}{(\sigma_X^2)^2} \left(\frac{\sigma_v^2}{n}\right) \end{aligned}$$

Rearranging terms yields the fundamental formula for the variance of the OLS estimator:

$$\boxed{\text{Var}(\hat{\beta}_1) = \frac{1}{n} \frac{\sigma_v^2}{(\sigma_X^2)^2} = \frac{\sigma_v^2}{n\sigma_X^4}}$$

where:

$$\sigma_v^2 = \text{Var}((X_i - \mu_X)u_i) \quad \text{and} \quad \sigma_X^2 = \text{Var}(X_i)$$

Combining this with asymptotic normality gives the **asymptotic sampling distribution of $\hat{\beta}_1$** :

$$\boxed{\hat{\beta}_1 \stackrel{a}{\sim} \mathcal{N}\left(\beta_1, \frac{1}{n} \frac{\sigma_v^2}{(\sigma_X^2)^2}\right)}$$

1.4 Homoskedasticity vs. Heteroskedasticity

1. The Homoskedastic Case (Special Case)

Errors are **homoskedastic** if the conditional variance of u_i given X_i is constant across all values of X_i :

$$\text{Var}(u_i | X_i) = \sigma_u^2 \quad \text{for all } i$$

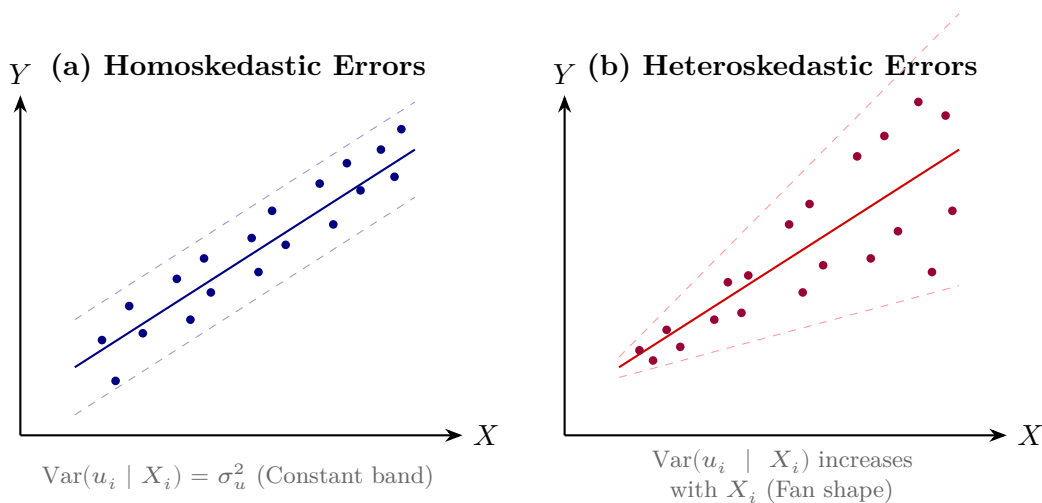


Figure 1: Homoskedastic vs. Heteroskedastic error distributions around the regression line.

Under homoskedasticity, we can simplify the numerator variance σ_v^2 :

$$\begin{aligned}
 \sigma_v^2 &= \mathbb{E}[(X_i - \mu_X)^2 u_i^2] \\
 &= \mathbb{E}[(X_i - \mu_X)^2 \mathbb{E}[u_i^2 | X_i]] \quad (\text{by LIE}) \\
 &= \mathbb{E}[(X_i - \mu_X)^2 \sigma_u^2] \quad (\text{since } \text{Var}(u_i | X_i) = \sigma_u^2) \\
 &= \sigma_u^2 \mathbb{E}[(X_i - \mu_X)^2] \\
 &= \sigma_u^2 \sigma_X^2
 \end{aligned}$$

Substituting $\sigma_v^2 = \sigma_u^2 \sigma_X^2$ into the general variance formula:

$$\text{Var}_{\text{homo}}(\hat{\beta}_1) = \frac{\sigma_u^2 \sigma_X^2}{n (\sigma_X^2)^2} = \frac{\sigma_u^2}{n \sigma_X^2}$$

$$\text{Var}_{\text{homo}}(\hat{\beta}_1) = \frac{\sigma_u^2}{n \sigma_X^2}$$

2. The Heteroskedastic Case (General Case)

Errors are **heteroskedastic** if the conditional variance depends on X_i :

$$\text{Var}(u_i | X_i) = \sigma_u^2(X_i) \neq \text{constant}$$

In this case, we cannot factor out σ_u^2 from the expectation, so $\sigma_v^2 \neq \sigma_u^2 \sigma_X^2$. We must retain the full

heteroskedasticity-robust variance formula:

$$\text{Var}_{\text{robust}}(\hat{\beta}_1) = \frac{1}{n} \frac{\sigma_v^2}{\sigma_X^4} = \frac{1}{n} \frac{\text{Var}((X_i - \mu_X)u_i)}{(\text{Var}(X_i))^2}$$

Estimating the Variances in Practice: Standard vs. Robust Estimators

The population formulas derived above depend on the **unobserved errors** u_i and the **unknown population moments** $\sigma_u^2, \sigma_v^2, \sigma_X^2$. In empirical applications, we estimate these variances by replacing u_i with the **OLS residuals** $\hat{u}_i = Y_i - \hat{Y}_i$ and population moments with their sample counterparts:

1. **Standard (Homoskedastic) Variance Estimator:** Assuming homoskedasticity, we estimate the error variance σ_u^2 using the squared **Standard Error of the Regression (SER)**:

$$s_u^2 \equiv \text{SER}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2 = \frac{\text{SSR}}{n-2}$$

Substituting s_u^2 into $\text{Var}_{\text{homo}}(\hat{\beta}_1) = \frac{\sigma_u^2}{\sum_{i=1}^n (X_i - \bar{X})^2}$ gives the **standard (classical) variance estimator**:

$$\widehat{\text{Var}}_{\text{homo}}(\hat{\beta}_1) = \frac{s_u^2}{\sum_{i=1}^n (X_i - \bar{X})^2} \Rightarrow \text{SE}_{\text{homo}}(\hat{\beta}_1) = \frac{s_u}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2}}$$

Limitation: This estimator is valid **only if** the homoskedasticity assumption holds. If errors are heteroskedastic, $\widehat{\text{Var}}_{\text{homo}}(\hat{\beta}_1)$ is biased and inconsistent, causing test statistics to be invalid.

2. **Heteroskedasticity-Robust Variance Estimator (Eicker-Huber-White / Sandwich Estimator):** Without assuming homoskedasticity, we directly estimate $\sigma_v^2 = \text{Var}((X_i - \mu_X)u_i)$ using the residuals $\hat{v}_i = (X_i - \bar{X})\hat{u}_i$. This yields the **heteroskedasticity-robust variance estimator**:

$$\widehat{\text{Var}}_{\text{robust}}(\hat{\beta}_1) = \frac{\sum_{i=1}^n (X_i - \bar{X})^2 \hat{u}_i^2}{[\sum_{i=1}^n (X_i - \bar{X})^2]^2} \Rightarrow \text{SE}_{\text{robust}}(\hat{\beta}_1) = \frac{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \hat{u}_i^2}}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

Advantage: This estimator is **consistent whether the errors are homoskedastic or heteroskedastic**. Because heteroskedasticity is common in economic data, modern empirical research almost universally defaults to robust standard errors.

Hypothesis Testing with the t -Statistic: With an estimator for the standard error $\text{SE}(\hat{\beta}_1)$ in hand (ideally heteroskedasticity-robust), we can now perform statistical inference as usual. Under the null hypothesis $H_0 : \beta_1 = \beta_{1,0}$ (most commonly $H_0 : \beta_1 = 0$),

the t -statistic:

$$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{\text{SE}(\hat{\beta}_1)} \stackrel{a}{\sim} \mathcal{N}(0, 1)$$

is asymptotically standard normal by the Central Limit Theorem. We can therefore carry out hypothesis tests (e.g., rejecting H_0 at the 5% significance level if $|t| > 1.96$), compute p -values, and construct asymptotic confidence intervals:

$$\hat{\beta}_1 \pm z_{1-\alpha/2} \times \text{SE}(\hat{\beta}_1)$$

in the standard fashion.

2 Omitted Variable Bias (OVB)

Least Squares Assumption 1 requires that the error term and the regressor are uncorrelated:

$$\text{Corr}(X_i, u_i) = 0 \quad (\text{or } \mathbb{E}[u_i | X_i] = 0)$$

When this assumption is violated, the OLS estimator is **biased and inconsistent**. The leading cause of this violation in observational data is **Omitted Variable Bias (OVB)**.

2.1 General Framework

Consider the linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

where u_i collects all other unobserved factors that influence Y_i .

If there is an omitted variable Z_i that affects Y_i , that variable is **hiding in the error term** u_i . Suppose we estimate the single-variable model of Y_i on X_i by OLS, ignoring Z_i .

Recall the formula for the OLS estimation error:

$$\hat{\beta}_1 - \beta_1 = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(u_i - \bar{u})}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

In large samples (by the Law of Large Numbers):

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(u_i - \bar{u}) \xrightarrow{p} \text{Cov}(X_i, u_i) = \sigma_{Xu}$$

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \xrightarrow{p} \text{Var}(X_i) = \sigma_X^2$$

Therefore:

$$\hat{\beta}_1 - \beta_1 \xrightarrow{p} \frac{\sigma_{Xu}}{\sigma_X^2}$$

Using the definition of correlation $\rho_{Xu} = \text{Corr}(X_i, u_i) = \frac{\sigma_{Xu}}{\sigma_X \sigma_u} \implies \sigma_{Xu} = \sigma_X \sigma_u \rho_{Xu}$, we rewrite the ratio:

$$\frac{\sigma_{Xu}}{\sigma_X^2} = \frac{\sigma_X \sigma_u \rho_{Xu}}{\sigma_X^2} = \left(\frac{\sigma_u}{\sigma_X} \right) \rho_{Xu}$$

This gives the fundamental **Omitted Variable Bias** formula:

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \left(\frac{\sigma_u}{\sigma_X} \right) \rho_{Xu}$$

2.2 The Two Conditions for Omitted Variable Bias

Because standard deviations are strictly positive ($\sigma_u > 0$ and $\sigma_X > 0$), $\hat{\beta}_1$ is biased and inconsistent whenever $\rho_{Xu} \neq 0$.

An omitted factor Z creates omitted variable bias if and only if **both** conditions hold:

1. **Z is a determinant of Y :** Z affects Y , meaning it is part of the error term u .
 2. **Z is correlated with X :** $\text{Corr}(X, Z) \neq 0$, which makes X correlated with the error term u ($\rho_{Xu} \neq 0$).
- If Z is not a determinant of Y , it does not enter u .
 - If Z is uncorrelated with X ($\text{Corr}(X, Z) = 0$), then $\rho_{Xu} = 0$, and $\hat{\beta}_1 \xrightarrow{p} \beta_1$ remains unbiased and consistent.

When both conditions hold, $\rho_{Xu} \neq 0$, and the sign of the bias is determined by the sign of ρ_{Xu} :

$$\text{sgn}(\text{Bias}) = \text{sgn}(\rho_{Xu})$$